Chapter 17 Problem Set

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1. Expanding $(5x+\frac{2}{x^2})^{24}$ using the Binomial Theorem, we get $\sum_{k=0}^{24}\binom{24}{k}(5x)^k\left(\frac{2}{x^2}\right)^{24-k}$. Simplifying that, we get $\sum_{k=0}^{24}\binom{24}{k}(5^k)(2^{24-k})x^{3k-48}$. We are interested in the part where $3k-48=3$, so $k=17$. Therefore, the coefficient is $\binom{24}{17}(5^{17})(2^7)$.

2. Let's consider the ratio $\frac{\binom{n}{k+1}}{\binom{n}{k}}$. Evaluating it, we get $\frac{n-k}{k+1}$. From this, we see that for $k<\frac{n}{2}$, $\frac{n-k}{k+1}>1$, so the coefficients are increasing. Also, for $k>\frac{n}{2}$, $\frac{n-k}{k+1}<1$, so the coefficients are decreasing. Therefore, the maximum is at $\binom{n}{\lfloor\frac{n}{2}\rfloor}$, so that is the largest coefficient of $(x+y)^n$.

We know that $(x+2y)^n=\sum_{k=0}^{n}2^k\binom{n}{k}x^ky^k$. Let's consider the ratio $\frac{2^{k+1}\binom{n}{k+1}}{2^k\binom{n}{k}}=\frac{2(n-k)}{k+1}$. Solving for $\frac{2(n-k)}{n+1}>1$, we get $k<\frac{1}{2}(n-1)$, so the coefficients are increasing until $k=\lfloor\frac{1}{2}(n-1)\rfloor$. Therefore, the largest coefficient is $\binom{n}{\lfloor\frac{1}{2}(n-1)\rfloor}$.

3. We see that $\frac{1}{k+1}\binom{n}{k}=\frac{n!}{(k+1)!(n-k)!}$, and we know $\binom{n+1}{k+1}=\frac{(n+1)!}{(k+1)!(n-k)!}$. Therefore, $\frac{1}{k+1}\binom{n}{k}=\frac{1}{n+1}\binom{n+1}{k+1}$. So $\sum_{k=0}^{n}\frac{1}{k+1}\binom{n}{k}=\frac{1}{n+1}\sum_{k=0}^{n}\binom{n+1}{k+1}=\frac{1}{n+1}(\sum_{k=0}^{n}\binom{n+1}{k}+\binom{n}{n}-\binom{n}{0})=\frac{2^{n+1}}{n+1}$.

4. We know that $(1+3)^n=\sum_{k=0}^{n}\binom{n}{k}3^k$ and that $(1-3)^n=\sum_{k=0}^{n}(-1)^k\binom{n}{k}3^k$. Subtracting both sums gives us the double desired sum. Therefore, the answer is $\frac{1}{2}(4^n-(-2)^n)=2^{2n-1}+(-2)^{n-1}$.

5. We know from before that $\sum_{0\le 2|k\le n}\binom{n}{k}x^k=\frac{1}{2}((1+x)^n+(1-x)^n)$. Substituting $x=1$, we get $\binom{n}{0}+\binom{n}{2}+\cdots=2^{n-1}$. Also, we know that $\sum_{0\le 4|k\le n}\binom{n}{k}=\frac{2^n+(1+i)^n+(1-i)^n}{4}$. Therefore, the answer we want is $2\sum_{0\le 4|k\le n}\binom{n}{k}-\sum_{0\le 2|k\le n}\binom{n}{k}=\frac{2^n+(1+i)^n+(1-i)^n}{2}-2^{n-1}=\frac{(1+i)^n+(1-i)^n}{2}$.

6. We know that $\sum_{k=0}^{2n}\binom{2n}{n}=2^{2n}=4^n$, so every component of that sum, including $\binom{2n}{n}$, must be less than 4^n. For the lower bound, we can rewrite $\frac{2^{2n-1}}{n}$ as $\frac{4^n}{2n}$, which is the average of all the components of the sum. Since $\binom{2n}{n}$ is the largest component, it is greater than the average.

8. Let's consider the number of k-dimensional faces of an n-dimensional cube. We can figure out the number of those k-dimensional faces based on the number of $k-1$ and k dimensional faces of an $(n-1)$-dimensional cube. Let's say $f(n,k)$ represents the number of k-dimensional faces of an n-dimensional cube. Then $f(n,k) = f(n-1,k-1) + 2f(n-1,k)$.

This can be thought of as extending the $(k-1)$-dimensional face into a new dimension and putting two k-dimensional faces on the ends. For example, you can create the twelve edges of a cube by extruding the four vertices and putting four edges each for the front and back, or you can create the six 2-dimensional faces of a cube by extruding the four edges of a square and putting two squares at the front and back.

We claim that $f(n,k) = \binom{n}{k}2^{n-k}$, and we can prove this by induction. First, we know that $f(0,0) = \binom{0}{0}2^{0-0} = 1$, because a 0-dimensional cube has one vertex. We also know that $f(1,0) = \binom{1}{0}2^{1-0} = 2$ because a one-dimensional cube has two vertices, and that $f(1,1) = \binom{1}{1}2^{1-1} = 1$ because a one-dimensional cube has one edge.

Now, let's assume $f(n,k) = \binom{n}{k}2^{n-k}$ and that $f(n,k+1) = \binom{n}{k+1}2^{n-k-1}$. Since $f(n,k) = f(n-1,k-1) + 2f(n-1,k)$, we know that $f(n+1,k+1) = f(n,k) + 2f(n,k+1) = \binom{n}{k}2^{n-k} + 2(\binom{n}{k+1}2^{n-k-1}) = 2^{n-k}\left(\binom{n}{k}\binom{n}{k+1}\right) = \binom{n+1}{k+1}2^{n-k}$, which completes the inductive step.

Now, since we know that $f(n,k) = \binom{n}{k}2^{n-k}$, we need to sum over all k to get the number of faces of an n-dimensional cube. This gives us $\sum_{k=0}^{n}\binom{n}{k}2^{n-k}$. By the Binomial Theorem, this is equal to $(1+2)^n = 3^n$. However, we don't want to count the face that represents the cube itself, so we subtract 1, giving us a final answer of $3^n - 1$.

9a. Consider the polynomial $f(x) = x^2 - x - 1$. One of its roots is $\alpha = \frac{1+\sqrt{5}}{2}$ and the other is $\beta = \frac{1-\sqrt{5}}{2}$. We know that $|\frac{1+\sqrt{5}}{2}| > 1$ and $|\frac{1-\sqrt{5}}{2}| < 1$, so $\frac{1+\sqrt{5}}{2}$ is a Pisot number.

Now, consider the polynomial $f(x) = x^2 - 2x - 1$. Its roots are $\alpha = 1+\sqrt{2}$ and $\beta = 1-\sqrt{2}$. We know that $|1+\sqrt{2}| > 1$ and $|1-\sqrt{2}| < 1$, so $1+\sqrt{2}$ is a Pisot number.

Finally, consider the polynomial $f(x) = x^2 - 3x - 1$. Its roots are $\alpha = \frac{3+\sqrt{13}}{2}$ and $\beta = \frac{3-\sqrt{13}}{2}$. We know that $|\alpha| > 1$ and $|\beta| < 1$, so $\frac{3+\sqrt{13}}{2}$ is a Pisot number.

9b. Since $\alpha = a + b\sqrt{d}$ is a Pisot number, $|a+b\sqrt{d}| > 1$ and $|a-b\sqrt{d}| < 1$, since its conjugate must be another root of the polynomial. The problem asks to prove that $\lim_{n\to\infty}\alpha^n = k$ for some integer k. By the Binomial Theorem,

$(a + b\sqrt{d})^n = \sum_{k=0}^{n} \binom{n}{k} a^k (b\sqrt{d})^{n-k}$. So

$$\lim_{n\to\infty} \alpha^n = \lim_{n\to\infty} \sum_{k=0}^{n} \binom{n}{k} a^k (b\sqrt{d})^{n-k} = \sum_{k=0}^{\infty} \binom{n}{k} a^k (b\sqrt{d})^{n-k}.$$

By the General Binomial Theorem, if $|a| < |b\sqrt{d}|$, then the sum converges.

I couldn't figure out how to prove that $|a| < |b\sqrt{d}|$, or how to prove that the sum converges to an integer instead of any real number.